

Omer Hayat

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EDUCATION

New York University Abu Dhabi

B.S. in Computer Science

Minors: Applied Mathematics, Philosophy, Artificial Intelligence

Abu Dhabi, UAE

Expected May 2028

EXPERIENCE

Robotics and Machine Learning Research Assistant

Aug. 2025 – Present

iCAS Lab, New York University Abu Dhabi

Abu Dhabi, UAE

- Designed a resource-aware semantic memory architecture for Unitree Go2 navigation, assigning continuous perception, memory updates, and safety validation to deterministic modules while reserving selectively invoked LLMs for higher-level semantic reasoning.
- Built a structured semantic memory and query engine that aggregates object-location observations and resolves natural-language targets, with explicit handling for ambiguous matches, unavailable objects, and unsafe destinations.
- Implemented a standalone ROS2 Foxy pipeline that reads semantic object-map data, validates candidate destinations against live occupancy grids, and publishes map-frame **PoseStamped** goals for Nav2 integration.
- Defined schema-validated interfaces and Dockerized evaluation tooling that decoupled perception, semantic memory, query interpretation, and navigation, enabling reproducible testing and independent integration across modules.

Adopt AI Fellow

May 2026 – Present

AdoptAI, The Lombard Hill Group

Remote

- Built a reusable Excel framework for AI adoption and economic assessment, automating three-year ROI, NPV, payback period, break-even, and benefit-cost calculations across 12 role-based worksheets covering product, workflow, user, platform, and organizational perspectives.
- Collaborating with the fellowship supervisor to redesign and develop AdoptAI's company website, translating business requirements into a clearer product narrative and modern user experience.

Software Engineer Intern

Feb. 2025 – Jun. 2025

Enrichly, Techstars '23

Abu Dhabi, UAE

- Designed and implemented responsive chatbot interfaces in Next.js, translating user feedback into iterative UI improvements, reusable components, and more intuitive conversational flows.
- Fine-tuned a Hugging Face language model for the company's conversational use case and integrated it into the chatbot's product workflow.
- Evaluated model responses for hallucinations, unsafe behavior, and use-case alignment through structured prompt testing and failure analysis, informing revisions to training data, prompts, and safeguards.

Data Analyst

Jul. 2023 – Aug. 2023

CardPay, NIC LUMS

Lahore, Pakistan

- Analyzed customer and market data using Python, SQL, and Tableau to identify and prioritize high-potential SME prospects, contributing to the acquisition of 6 new clients during a two-month internship.

PROJECTS

PitchMirror | *Next.js, Fastify, FFmpeg, AWS, Amazon Bedrock, Amazon Nova Pro*

- Built and deployed a multimodal presentation-coaching platform that processes video metadata, representative keyframes, and optional transcripts into structured feedback across voice, presence, and content.
- Engineered an asynchronous AWS pipeline using S3, DynamoDB, Step Functions, ECS Fargate, and Bedrock, with idempotent job tracking, schema-validated AI outputs, automated repair, and deterministic fallbacks for reliable report generation.

RepoTriage | *Python, FastAPI, scikit-learn, SQLAlchemy, PostgreSQL, Next.js, Docker*

- Built an end-to-end GitHub issue intelligence system for **pandas-dev/pandas** that predicts likely issue labels, abstains on low-confidence cases, retrieves similar historical issues, and records maintainer feedback.
- Developed reproducible ML and serving pipelines with temporal train/validation/test splits, calibrated confidence thresholds, deterministic TF-IDF cosine retrieval, FastAPI inference and feedback endpoints, SQLAlchemy persistence, and automated tests.

Kaboom: Ray-Marched Fireball Renderer | *C++, OpenMP*

- Built a dependency-free CPU ray-marching renderer that generated a 120-frame, 640×480 animated fireball using signed distance fields, conservative sphere tracing, and sinusoidal and fractal Brownian motion displacement.
- Implemented procedural value noise, finite-difference surface normals, Lambertian and rim lighting, fire-palette interpolation, and OpenMP-parallelized frame rendering, exporting binary PPM frames for GIF and MP4 generation.

TECHNICAL SKILLS

Languages: Python, C++, TypeScript/JavaScript, SQL, R, HTML/CSS

Machine Learning & Data: scikit-learn, PyTorch, TensorFlow, Hugging Face, pandas, NumPy, Matplotlib

Frameworks & Systems: ROS2, FastAPI, SQLAlchemy, Fastify, Next.js, React, FFmpeg, PostgreSQL

Cloud & Developer Tools: AWS (S3, DynamoDB, Step Functions, ECS Fargate, Bedrock), Docker, Linux, CUDA, Git

Certifications: DeepLearning.AI Machine Learning Specialization; Google Data Analytics Professional Certificate